

AI-Assisted ICU Early Warning System Using VitalDB Physiological Waveforms

1. Introduction

Continuous physiological monitoring is a critical component of modern intensive care and perioperative healthcare systems. Traditional monitoring systems generally rely on static threshold-based alarms to detect physiological abnormalities such as hypotension, oxygen desaturation, and tachycardia. However, these alarm systems frequently generate false alerts and often fail to recognize subtle physiological deterioration patterns before a critical event occurs.

The objective of this work is to investigate whether machine learning techniques can identify physiological instability patterns from continuously monitored waveform and vital sign data. The study utilizes the VitalDB dataset, a large-scale physiological waveform repository containing real-time monitoring signals collected during clinical procedures.

The developed framework focuses on detecting:

- Hypotension events
- Oxygen desaturation events
- Tachycardia events

using machine learning-based physiological analysis.

2. Objectives

The primary objectives of this study are:

1. To acquire and preprocess physiological waveform data from the VitalDB dataset.
 2. To perform exploratory data analysis on vital sign signals.
 3. To extract meaningful physiological features from ECG, MAP, HR, and SPO2 signals.
 4. To generate clinically relevant deterioration labels.
 5. To train machine learning models for physiological event prediction.
 6. To compare multiple machine learning algorithms.
 7. To build a risk-scoring framework for early physiological deterioration detection.
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3. Dataset Description

Dataset

VitalDB Physiological Monitoring Dataset

Signals Selected

Signal	Description
ECG	Electrocardiogram waveform
HR	Heart Rate
MAP	Mean Arterial Pressure
SPO2	Oxygen Saturation

Initial Cohort

A cohort containing all required physiological signals was selected using the VitalDB API.

Final Dataset

- Total Cases Processed: 96
 - Total Temporal Windows: 1175
 - Total Features: 26
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4. Methodology

Phase 1: Data Acquisition

The VitalDB Python API was used to retrieve waveform recordings containing:

- ECG
- HR
- MAP
- SPO2

Each case was downloaded and stored as a structured CSV file.

Phase 2: Data Preprocessing

Several preprocessing steps were applied.

Missing Value Handling

Missing observations were handled using:

- Forward Fill
- Backward Fill
- Interpolation

Signal Validation

Physiological constraints were applied:

Heart Rate

$$20 \leq \text{HR} \leq 250$$

Mean Arterial Pressure

$$40 \leq \text{MAP} \leq 200$$

Oxygen Saturation

$$50 \leq \text{SPO}_2 \leq 100$$

Invalid values were removed.

Noise Removal

Incomplete records and unusable signals were discarded.

Phase 3: Exploratory Data Analysis

EDA was conducted for each physiological signal.

Analyses Performed

- Distribution Analysis
- Histograms
- Correlation Analysis
- Missing Value Analysis
- Signal Trend Visualization

Findings

- MAP variability was observed across multiple cases.
 - SPO2 remained relatively stable in most cases.
 - ECG waveforms demonstrated varying physiological patterns.
 - HR fluctuations indicated different cardiovascular states.
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Phase 4: Window Segmentation

Continuous signals were segmented into fixed-size windows.

Window Configuration

- Window Size: 3000 samples

- Step Size: 1000 samples

This approach generated multiple physiological samples from each case and increased the number of training observations.

Phase 5: Feature Engineering

Basic Features

For each signal:

- Mean
- Standard Deviation
- Minimum
- Maximum

MAP Features

- Mean MAP
- MAP Range
- MAP Variability

SPO2 Features

- Mean SPO2
- SPO2 Drop

HR Features

- Mean HR
- HR Variability

ECG Features

- Mean ECG
 - ECG Variance
 - ECG Range
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Phase 6: Advanced Feature Engineering

Additional physiological indicators were derived.

MAP Variability

Represents blood pressure instability.

HR Variability

Represents cardiovascular fluctuation.

ECG Range

Captures waveform amplitude variation.

MAP Drop

Measures severity of pressure reduction.

SPO2 Drop

Measures oxygen instability.

Phase 7: Label Generation

Three prediction targets were generated.

Hypotension

Label = 1 if:

MAP < 65

Otherwise:

Label = 0

Tachycardia

Label = 1 if:

HR > 100

Otherwise:

Label = 0

Oxygen Desaturation

Label = 1 if:

Minimum SPO2 < 94

Otherwise:

Label = 0

5. Machine Learning Pipeline

Data Split

Train-Test Split:

- Training Data: 80%
- Testing Data: 20%

Stratified sampling was used.

Imbalance Handling

SMOTE was applied to:

- Hypotension
- Tachycardia

to address severe class imbalance.

Feature Scaling

StandardScaler was used for:

- Logistic Regression
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Models Evaluated

Logistic Regression

Baseline linear model.

Random Forest

Ensemble tree-based model.

XGBoost

Gradient boosting model.

6. Results

Hypotension Prediction

Model	Accuracy	Precision	Recall	F1	ROC-AUC
Logistic Regression	0.9915	0.7500	1.0000	0.8571	0.9993
Random Forest	1.0000	1.0000	1.0000	1.0000	1.0000
XGBoost	1.0000	1.0000	1.0000	1.0000	1.0000

Low SPO2 Prediction

Model	Accuracy	Precision	Recall	F1	ROC-AUC
Logistic Regression	0.9957	0.9796	1.0000	0.9897	1.0000
Random Forest	1.0000	1.0000	1.0000	1.0000	1.0000
XGBoost	1.0000	1.0000	1.0000	1.0000	1.0000

Tachycardia Prediction

Model	Accuracy	Precision	Recall	F1	ROC-AUC
Logistic Regression	0.9872	0.6667	1.0000	0.8000	1.0000
Random Forest	1.0000	1.0000	1.0000	1.0000	1.0000
XGBoost	1.0000	1.0000	1.0000	1.0000	1.0000

7. Discussion

The results demonstrate that physiological waveform-derived features contain significant predictive information.

Tree-based models consistently outperformed Logistic Regression due to their ability to capture nonlinear relationships within physiological data.

Random Forest and XGBoost achieved perfect performance on all three prediction tasks.

However, the extremely high performance suggests possible feature-label leakage because some labels were directly derived from physiological measurements used as model features.

A secondary leakage-free experiment is recommended to further validate generalization capability.

8. Clinical Findings

1. MAP-related features strongly influence hypotension prediction.
 2. SPO2 instability features contribute significantly to oxygen desaturation detection.
 3. Heart-rate variability contributes to tachycardia detection.
 4. Physiological waveform information provides useful indicators of patient deterioration.
 5. Machine learning models outperform simple threshold-based monitoring strategies.
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9. Limitations

1. Labels were generated from existing physiological variables.
 2. Potential feature-label leakage may inflate model performance.
 3. Only VitalDB data was used in this module.
 4. Deep learning models such as LSTM and CNN were not evaluated.
 5. External validation on independent datasets was not performed.
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10. Future Work

1. Perform leakage-free experiments.
 2. Integrate MIMIC-IV and eICU datasets.
 3. Develop real-time streaming prediction.
 4. Implement LSTM-based temporal modeling.
 5. Build multimodal ensemble prediction frameworks.
 6. Deploy a real-time ICU risk scoring dashboard.
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11. Conclusion

A complete physiological monitoring and machine learning pipeline was successfully developed using the VitalDB dataset. The framework includes data acquisition, preprocessing, exploratory analysis, feature engineering, label generation, model training, evaluation, and risk assessment. The results demonstrate that physiological waveform-derived features can effectively identify clinically relevant deterioration events such as hypotension, oxygen desaturation, and tachycardia. The developed framework serves as a foundational component of an AI-assisted ICU early warning system.